

Springs, Rods, and Ropes

It is at this time that i decided to make my notes into a comprehensive textbook, so things before this are incomplete. My apologies to the reader.

Recall:

Last class we talked about isolation diagrams.

Choose a coordinate system, making sure it is inertial
write

$$\vec{f}_{net \rightarrow i}^{ext} = M_i \vec{a}_{i,cm}$$

Force relationships

$$F_{grav} = mg \quad (1)$$

$$F_r = k\Delta x \quad (2)$$

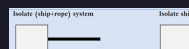
If we assume a string to be massless, we can just do forces on each side.

E.g. Spaceship towing

The Star Wars ship Enterprise of mass M is stranded in space.
A friendly ship pulls a **rope with mass m** at constant force F_0 .
(a) What is the acceleration of the Enterprise as it is towed?
(b) What is the contact force the rope applies to the Enterprise?



The rope has mass, so the force has to be transmitted along the rope. The rope has to carry itself. The force from the right ship is different from the force on the left ship (?). We can isolate any combination of systems. For instance, if we isolate the ship+rope system



We have force F_0 pulling on the rope.

We have an acceleration to the right. We expect acceleration to have magnitude $|a|$ and point to the right.

We can say that $F_{net}^{ext} = M_{tot} A_{cm,x}$

$F_0 = (M + m)a$ which we can easily rewrite $a = \frac{F_0}{M+m}$

With the system as the combined masses made the acceleration easy to find, so we don't need to worry about the internal contact forces. We still have to solve for the contact force though.

1 if we choose to isolate the ship

the rope has a Force on the right of the rope $T_r = F_0$. The rope also has a tension on the left $T_l = T_c$ the force on the right of the ship.

We can write that the total force

$$T_r - T_l = F_{tot} = ma$$

$$\text{and } T_l = Ma = \frac{MF_0}{M+m}$$

we can rewrite the expression with

$$F_0 - \left(\frac{M}{m+M}\right)F_0 = \left(\frac{m}{m+M}\right)F_0 \quad (3)$$

$$0 = 0 \quad (4)$$

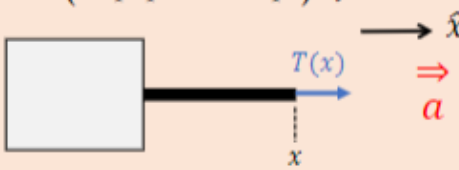
so we know that our math checks out.

We can check the units. In the massless rope case, we get tension on the left and right as the same. This fits with our gut check.

What we just found, though, was that the tension on the left side of the rope is some fraction of the tension on the rope on the right.

If we have a uniform rope, we can split the rope anywhere and find the tension on the right and left of that point. We can find the tension as a function as how far along the rope we are.

Isolate (ship+partial-rope) system

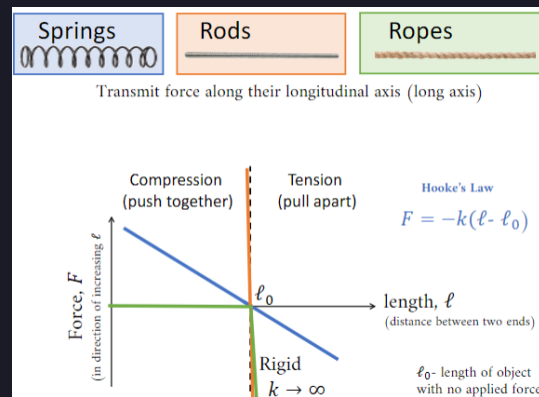


$$F_{NET,x}^{ext} = M_{tot} a_x$$

$$T(x) = \left(M + m \frac{x}{L}\right) a$$

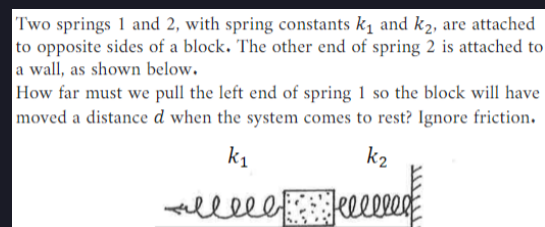
$$T(x) = \left(M + m \frac{x}{L}\right) \frac{F_0}{M+m}$$

How does this change if we allow the rope to stretch with applied force? what about a spring?

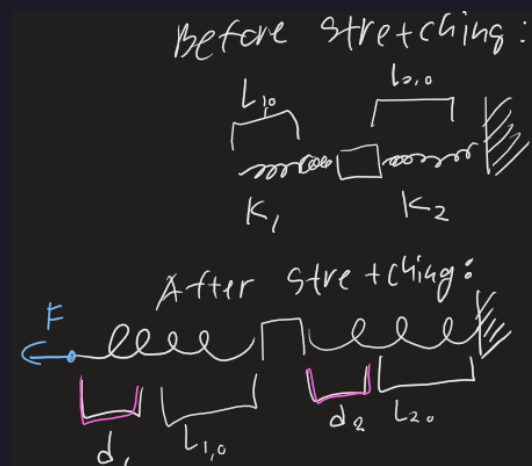


How steep this line is tells us the stiffness of the spring, small slope is easy to push and pull the spring.

Ropes don't provide any force under compression, but are rigid in tension.



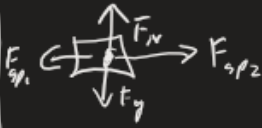
Lets draw a picture to more clearly see what's happening.



Total displacement left is $\Delta x = d_1 + d_2$

To relate how the spring act, we can isolate the block.

The springs are under tension and want to pull the block. The block is stationary, so $\vec{a} = 0$.



$$F_{\text{net}} = M \vec{a} = 0$$

$$F_{s2} - F_{s1} = 0$$

Using Hooke's law, $|F_{s1}| = |k_1 x|$

$$k_2 d_2 = k_1 d_1$$

$$d_1 = \frac{k_2}{k_1} d_2$$

$$\Delta x = \frac{k_2 d_1}{k_1} + d_1$$

$$= \left(\frac{k_2}{k_1} + 1 \right) d_1$$

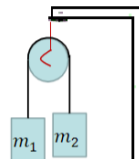
We can check this. If we make k_1 really rigid and take $\lim_{k_1 \rightarrow \infty}$, the displacement becomes only d_1 .

Applying constraints: Atwood's Machine

Two blocks are attached to either end of a massless string, which is draped over a frictionless peg. The peg is hung from a stand as shown in the picture, with $m_2 > m_1$.

(A) Just after the blocks are released from rest, what is the acceleration of each block?

(B) What is the magnitude F_0 of the force that holds up the peg?



Isolate m_1 , Isolate m_2

Unknowns: T, a_1, a_2 .

Because the rope can't stretch, $a_1 = -a_2$.

the length of the rope is constant
 $= (y_p - y_1) + (y_p - y_2) + (\pi R)$
 $= 2y_p - y_1 - y_2 - \pi R$
 $\frac{d}{dt}L = 0 = v_{1,y} - v_{2,y}$
 $v_{1,y} = -v_{2,y}$
 + we get acceleration,
 $a_{1,y} = -a_{2,y}$

this is what we actually need
 we just didn't want to rely on our
 intuition.

① $T - m_1 g = m_1 a$
 ② $m_2 g - T = m_2 a$

$T - m_1 g + m_2 g - T = m_1 a + m_2 a$
 $(m_2 - m_1)g = (m_1 + m_2)a$
 $a = \frac{(m_2 - m_1)}{m_1 + m_2} g$

We can take this exact process for a far more complicated system.

A more complex constraint

A system is hung from the ceiling as shown in the diagram. Two massless and frictionless pegs are freely suspended and have a massless rope that winds around them. From Peg 1, a mass m_1 is rigidly attached to its center. Peg 2 is attached to the ceiling by a spring that connects to its center.

What is the acceleration of Peg 2?

See next: *Forces inside an object*

See previous: *Gravity and electromagnetism*